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EXPERIMENT 07
Integration with Zoho CRM using APIs

Aim

Sync enquiry contacts to Zoho CRM as leads with an authenticated connection and retry-safe handling.

Procedure

1. Use a test CRM organization approved by your institution. Send only student name and email, not marks, date of birth or message.
2. Create a Creator connection admission_crm with the scope ZohoCRM.modules.leads.WRITE. Never paste raw tokens into scripts.
3. Use Email as the duplicate-check key. Map Student_Name to Last_Name, Email to Email, returned ID to CRM_Lead_ID.
4. Mark a record Pending after save; call the upsert after save or from a scheduled queue so admission handling never depends on CRM being up.
5. Check both HTTP errors and the record-level status in the response before setting Synced; on error store a sanitized message.
6. Re-run the same request and confirm only one lead exists.

```
lead = Map(); lead.put("Last_Name", rec.Student_Name); lead.put("Email", rec.Email);
payload = Map(); payload.put("data", {lead}); payload.put("duplicate_check_fields", {"Email"});
resp = invokeurl [ url :"https://www.zohoapis.com/crm/v8/Leads/upsert"
    type :POST parameters:payload.toString() connection:"admission_crm" ];
```

Use the API domain for your own data center. Add response and exception handling before enabling retries. If Student_Name is a Name-type field, map its text value explicitly.

Implementation and verification

Test	Expected evidence
Valid enquiry sync	One lead, ID stored, status Synced
Repeat request	No duplicate
Invalid email	Readable error, not Synced
Revoked connection	Authorization failure logged, no secrets
CRM outage	Retry or Pending; admission flow unaffected

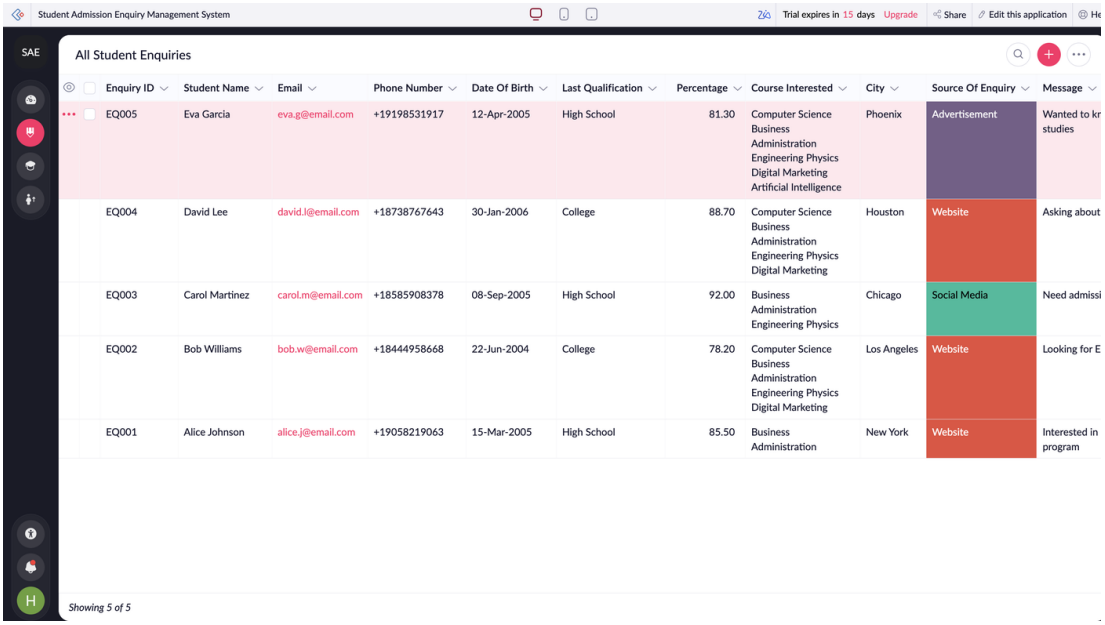
Field mapping

Creator field	CRM field	Sent?
Student Name	Last_Name	Yes
Email	Email	Yes (duplicate key)
CRM_Lead_ID (new)	id (returned)	Stored on success only
Percentage, Date Of Birth, Phone, City, Message	-	No (data minimization)

Evidence and testing

E07: field mapping, redacted connection config, sanitized response, CRM verification, retry log. Fields to add to the form: CRM_Lead_ID, CRM_Sync_Status (Pending / Synced / Error), CRM_Error_Message. Record results after running the tests above; none are claimed here.

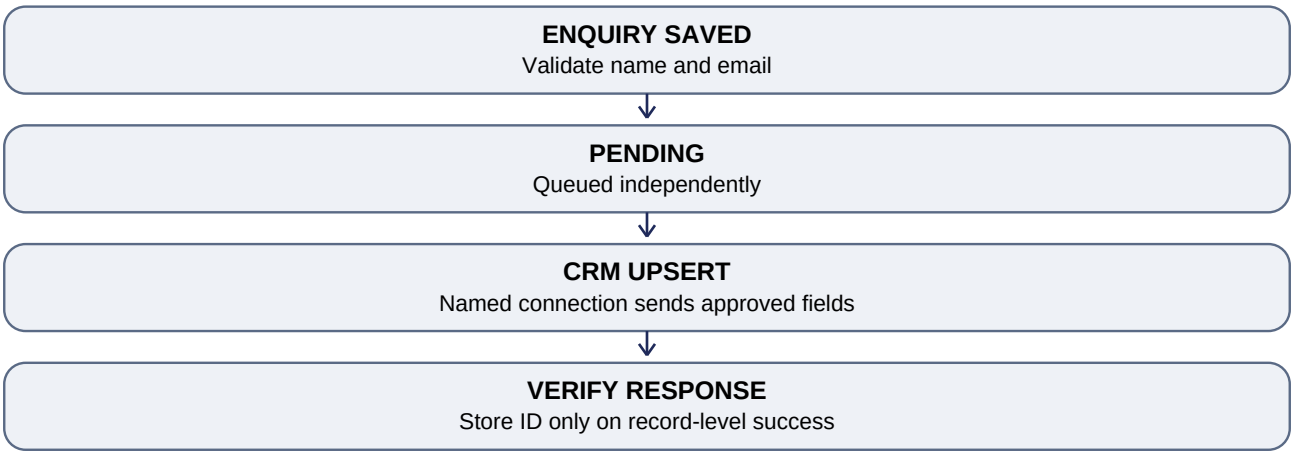
Application images and explanation



Enquiry ID	Student Name	Email	Phone Number	Date Of Birth	Last Qualification	Percentage	Course Interested	City	Source Of Enquiry	Message
EQ005	Eva Garcia	eva.g@email.com	+19198531917	12-Apr-2005	High School	81.30	Computer Science Business Administration Engineering Physics Digital Marketing Artificial Intelligence	Phoenix	Advertisement	Wanted to kr studies
EQ004	David Lee	david.l@email.com	+18738767643	30-Jan-2006	College	88.70	Computer Science Business Administration Engineering Physics Digital Marketing	Houston	Website	Asking about
EQ003	Carol Martinez	carol.m@email.com	+18585908378	08-Sep-2005	High School	92.00	Business Administration Engineering Physics	Chicago	Social Media	Need admissi
EQ002	Bob Williams	bob.w@email.com	+18444958668	22-Jun-2004	College	78.20	Computer Science Business Administration Engineering Physics Digital Marketing	Los Angeles	Website	Looking for E
EQ001	Alice Johnson	alice.j@email.com	+19058219063	15-Mar-2005	High School	85.50	Business Administration	New York	Website	Interested in program

Figure 1: All Student Enquiries report - source records (name and email only are synced)

Process diagram



Generated explanatory diagram. It is not a screenshot of an executed workflow.

Result

Expected result and verification criteria. A test enquiry syncs once, repeats do not duplicate, and failures are visible and recoverable.